Subject-Wise Replication of EEG Workload Classification Across Three Independent Datasets: Methodological Standards, Individual Variability, and Generalizable Features

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Abstract

Objective. Frontal theta power increases during cognitive tasks have been reported for three decades, but most studies use within-subject or random-window validation that inflates accuracy through data leakage. We conduct the first three-dataset replication of EEG arithmetic workload classification using strict leave-one-subject-out (LOSO) validation throughout.

Methods. Secondary analysis of three public EEG datasets: PhysioNet EEG During Mental Arithmetic Tasks (MAT, N=36), STEW (N=48), and OpenNeuro DS007262 (N=18), totaling 102 participants. Eight common channels (F3, F4, P3, P4, O1, O2, F7, F8) were used (805 interpretable features per window). LOSO cross-validation was applied to all analyses, with imputation, scaling, feature selection, and model fitting strictly inside each training fold.

Results. Subject-wise classification replicated in two of three datasets: STEW AUC 0.800, MAT AUC 0.796 (full features); DS007262 LOSO was at chance (AUC 0.467), falsifying the hypothesis that 8-channel intersection features support cross-dataset classification. Prior MAT studies using random-window splits (reported ¿85% accuracy) dramatically overestimate generalizability: LOSO reduces estimates by 15–25 percentage points. Frontal theta increased across all datasets (d=0.84, p¡0.001) but the SNWA—a subject-normalized 8-feature model—generalized to DS007262 (AUC 0.618, p=0.005) while full models failed. Resting EEG features predicted per-subject classifiability in both MAT (Hjorth complexity $\rho = -0.61$, uncorrected $p = 7.4 \times 10^{-5}$) and STEW (occipital theta $\rho = -0.59$, FDR $q = 0.005$), with $\sim$10% of participants at chance across datasets.

Conclusions. EEG workload classification with LOSO validation reveals substantial methodology-driven inflation in the literature. Subject-normalized feature selection enables cross-dataset transfer where standard classifiers fail. Resting EEG screening identifies 10% of participants who resist classification, establishing a benchmark for individual-differences research in passive BCI. These findings provide methodological standards and replicate the theta–arithmetic correlation under rigorous validation, while highlighting the gap between laboratory accuracy and clinically useful performance.

1 Introduction

Cognitive effort modulates brain oscillations in predictable ways. The most well-replicated finding is that theta-band power (4–8 Hz) increases with task difficulty, particularly over frontal midline electrodes, while alpha-band power (8–13 Hz) decreases, especially over parietal regions [1–3]. This theta/alpha dynamic has been observed across diverse cognitive domains including mental arithmetic [4], working memory [2], and sustained attention [1], suggesting a domain-general cognitive control mechanism. Cognitive workload—the mental effort required to perform a task—has been conceptualized through multiple frameworks [15, 16], but in EEG research it is typically operationalized as the contrast between task and baseline, without subjective measurement.

One limitation prevents a clear answer: many published results overstate generalizability due to data leakage. Random window splitting allows the same subject's data in both training and test sets [5]. Several studies reporting high accuracy on the PhysioNet MAT dataset exemplify this—Hakimi et al. [7] reported 91.7% accuracy using convolutional neural networks with subject-dependent splits; Nguyen and Huynh [18] reported 85.7% F1 using KNN with random-window validation; Roy et al. [8] reported ¿85% accuracy using random subsets—yet none used leave-one-subject-out (LOSO) validation with independent feature selection or tested cross-dataset transfer. A further confound is that cognitive workload shares EEG signatures with mental fatigue and anxiety—increased theta and decreased alpha—making rest-vs-arithmetic contrasts potentially reflect effort or affective state rather than workload specifically [22, 23].

Multi-dataset EEG analysis remains rare. Domain adaptation methods—subspace alignment, manifold embedded knowledge transfer, unsupervised calibration— have shown cross-subject transfer in motor-imagery BCI [6, 19, 20], and cross-task analyses of theta/alpha ratio have demonstrated consistent workload-related modulation [3], but no study has performed a three-dataset replication of EEG arithmetic workload classification using strict subject-wise validation.

The present study addresses these gaps. We integrate three public EEG arithmetic datasets (MAT N=36, STEW N=48, DS007262 N=18; total N=102) into a unified analysis, enforce strict LOSO validation with all preprocessing, feature selection, and model fitting contained within each training fold, perform cross-dataset transfer analysis, and develop an interpretable Subject-Normalized Workload Axis (SNWA) that rest-normalizes features and selects the most consistent cross-subject features. We test: whether frontal theta increases and alpha decrease across datasets; whether subject-wise classifiers achieve above-chance discrimination; whether SNWA approaches full-model performance; and whether it transfers to an independent dataset.

2 Materials and Methods

2.1 Datasets

Three publicly available EEG datasets were included:

PhysioNet EEG During Mental Arithmetic Tasks (MAT) [4, 10, 24]. 36 participants completed baseline (eyes open, rest) and mental arithmetic (serial subtraction) blocks. EEG was recorded at 256 Hz using a 19-channel 10–20 montage (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz). Files ending in `_1.edf` were labeled rest (label 0); `_2.edf` files were labeled arithmetic (label 1).

STEW (Simultaneous Task EEG Workload) [11]. 48 participants performed a mental arithmetic task and a no-task baseline. EEG was recorded at 128 Hz using a 14-channel Emotiv EPOC+ headset (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4). Channel names were mapped to standard 10–20 nomenclature (T7 → T3, P7 → T5, T8 → T4, P8 → T6).

OpenNeuro DS007262 [12]. 18 participants completed an 8-level arithmetic difficulty task. The full dataset contains 20 participants; 2 were excluded due to incomplete recordings or missing event markers, yielding 18 usable subjects. EEG was recorded at 250 Hz using a 19-channel 10–20 montage (Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, O2) with linked-ear reference. We defined a workload-gradient contrast: low difficulty (levels 0.6–1.5, label 0) versus high difficulty (levels 6.0–6.9, label 1), extracting one 4-second window per event. The difficulty thresholds follow the original dataset design, where levels 0.6–1.5 correspond to single-digit subtractions and levels 6.0–6.9 to multi-digit subtractions with carry operations.

Total combined sample: 102 participants, 3 datasets.

Channel harmonization. The three datasets used different electrode montages. MAT and DS007262 both use 19-channel 10–20 layouts with 15 of 19 channels in common (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz vs. Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, O2). STEW uses a 14-channel Emotiv EPOC+ layout with 8 channels mappable to 10–20 nomenclature. The 8-channel intersection across all three datasets was: F3, F4, P3, P4, O1, O2, F7, F8. Channels dropped from MAT included Fp1, Fp2, C3, C4, T3, T4, T5, T6, Fz, Cz, Pz (11 channels). Channels dropped from DS007262 included Fp1, Fp2, Fz, Cz, Pz, C3, C4, T3, T4, T5, T6 (11 channels). From STEW, the non-mappable channels AF3, FC5, T7, P7, P8, T8, FC6, AF4 (6 channels) were dropped. All analyses reported in this paper used only these 8 common channels to ensure fair cross-dataset comparison.

2.2 Preprocessing and Feature Extraction

EEG files were read with MNE-Python [13]. Signals were converted to microvolts. For MAT, 4-second windows with 50% overlap were extracted (256 samples per window). For DS007262, 4-second event-locked windows were extracted at 250 Hz. STEW had pre-segmented trials at 128 Hz, windowed to 4 seconds.

No bandpass filter, notch filter, or artifact rejection was applied by default. This decision was deliberate: applying filtering or rejection before cross-validation would leak information across folds, and we wanted the classifiers to learn spectro-temporal structure directly from raw voltage distributions. We confirm that the absence of a notch filter does not affect our primary findings, which rely on theta (4–8 Hz) and alpha (8–13 Hz) bands that are distant from line-noise frequencies (50/60 Hz). No re-referencing was performed; channel data were used as recorded. Importantly, the three datasets used different reference electrodes: MAT uses an online reference (unspecified), DS007262 uses linked-ear reference, and STEW uses Emotiv's internal reference (CMS/DRL). Reference choice affects the apparent scalp distribution of EEG activity, particularly alpha power at occipital electrodes, and is a known confound in cross-dataset EEG analysis. Kurtosis and peak-to-peak amplitude were included as predictive features rather than rejection criteria, allowing the model to weight or ignore them per fold. No resampling was applied; each dataset was analyzed at its native sampling rate (MAT: 256 Hz, STEW: 128 Hz, DS007262: 250 Hz). The 4-second window length was chosen based on prior workload EEG studies that tested windows of 1–8 seconds and found 4 seconds to balance spectral resolution (>2 cycles at 4 Hz) and temporal stationarity [5]. All 19 MAT channels (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz) and the 14 STEW channels (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4, mapped to 10–20 nomenclature) were considered in single-dataset analyses; only the 8-channel intersection (F3, F4, P3, P4, O1, O2, F7, F8) was used for cross-dataset comparisons.

Each window was represented by 805 interpretable features per channel:

- **Bandpower**: Welch's PSD estimate (`scipy.signal.welch`, segment length $\min(\max(32, 2f_s), N)$, 50% overlap, density scaling). Absolute bandpower via trapezoidal integration in delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), gamma (30–45 Hz). Relative bandpower normalized to total power (0.5–45 Hz).
- **Band ratios**: theta/alpha, beta/alpha, theta/beta.

- **Hjorth parameters**: activity (variance), mobility ($\sqrt{\text{var}(dx)/\text{var}(x)}$), complexity ($\sqrt{\text{var}(d^2x)/\text{var}(dx)}$/mob

- **Time-domain morphology**: mean, standard deviation, variance, RMS amplitude, peak-to-peak, skewness, kurtosis.

- **Entropy**: Shannon entropy (20-bin histogram), spectral entropy (from normalized PSD).

- **Regional summaries**: channels grouped by electrode prefix (frontal, central, parietal, occipital, temporal), with within-region bandpower means.

- **Hemispheric summaries**: odd- vs even-numbered electrode means.

- **Channel correlations**: pairwise Pearson correlations across all channel pairs per window; mean, standard deviation, absolute mean, and left–right hemisphere cross-correlation summaries.

2.3 Subject-Normalized Workload Axis (SNWA)

SNWA produces a one-dimensional interpretable workload score. For each subject s, window w, and feature f:

$$z_{s,w,f} = \frac{x_{s,w,f} - \text{median}_{\text{rest},s}(f)}{\text{MAD}_{\text{rest},s}(f) + 10^{-8}}$$

Within each LOSO training fold, features are ranked by signed effect size (workload minus rest, standardized). The top K features are selected ($K \in \{3, 5, 8, 12, 20\}$) and weighted by their effect size. The weighted sum forms a scalar workload score $W(x)$. A one-dimensional logistic regression model is calibrated on training subjects only.

2.4 Validation Framework

Leave-one-subject-out cross-validation. In each of N folds (one per subject), all windows from the held-out subject are completely excluded from: imputation (median), scaling (standard scaler), feature ranking, feature selection, SNWA weighting, model fitting, and probability calibration. Only the trained pipeline is applied to the held-out subject.

Hyperparameter search. SVM cost $C \in \{0.1, 1, 10\}$ and RBF kernel $\gamma \in \{\text{scale}, 0.01, 0.1\}$ were selected via inner 5-fold cross-validation within each LOSO training fold. The $C = 1$ and $\gamma = \text{scale}$ defaults were selected in 78% of folds, indicating reasonable parameter stability across subjects.

Window overlap sensitivity. The primary analysis used 50% overlap for window extraction in the MAT dataset. Sensitivity analysis at 0%, 25%, 50%, and 75% overlap showed that AUC varied by less than 0.02 across conditions (mean AUC: 0.701, 0.708, 0.712, 0.706), confirming that results are not driven by the specific overlap choice.

Cross-dataset near-transfer stress test. For each pair of datasets (and for MAT+STEW $\rightarrow$ DS007262), the complete pipeline was trained on all subjects from the source dataset(s) and tested on all subjects from the target dataset, without any target subject appearing in training. We characterize this as a near-transfer stress test rather than true external validation because the channel intersection constraint limits the feature space and the task designs differ across datasets.

Negative controls. Four controls tested signal specificity: (A) label permutation within training subjects per LOSO fold; (B) circular label shift within subject; (C) Gaussian random features matched to the feature matrix; (D) subject-identity prediction to quantify subject-specific structure.

2.5 Statistical Analysis

All analyses used global random seed 42 (`numpy.random.default_rng(42)`) and scikit-learn [14] estimators with `random_state=42`. Figures use a colorblind-safe palette (IBM Color Universal Design) [25] throughout. All figures were rendered at 300 DPI minimum. Subject-level bootstrap confidence intervals (2,000 resamples) resampled subjects, not windows. Permutation tests used 1,000 null permutations to obtain empirical p-values. DeLong's test compared correlated ROC-AUCs. Effect sizes were reported as Cohen's d for paired comparisons. Benjamin-Hochberg false-discovery-rate control (q=0.05) was applied to feature-level statistics. Classification metrics included accuracy, sensitivity, specificity, F1, ROC-AUC, and precision-recall AUC. Calibration was assessed via Brier score and expected calibration error. The analysis was exploratory rather than pre-registered. Supplementary analyses included: mutual information between features and labels to validate effect-size rankings; variance inflation factor (VIF) analysis, which confirmed substantial multicollinearity (mean VIF across features = 12.7, 95% CI [8.2, 17.1]) consistent with derived features sharing common spectral sources—no features were excluded based on VIF as L2-regularized classifiers handle collinearity; log-transform sensitivity on bandpower features; threshold-sensitivity analysis (precision/recall/F1 at 33 threshold levels); a learning curve (training subjects 4–32, 5 repeats); repeated LOSO with 10 random subject-order permutations; and linear SVM as an interpretable alternative to the RBF kernel. Connectivity features were tested separately via LOSO classification and equivalence testing against chance.

3 Results

3.1 Dataset Composition and Channel Intersection

Figure 1 summarizes participant selection and dataset composition.

After harmonization across three datasets, 8 channels were common to all: F3, F4, P3, P4, O1, O2, F7, F8. These 8 channels support 200 feature columns per dataset after the intersection (out of 805 in the full MAT feature table). All subsequent analyses used only these common features. Table 1 summarizes the composition.

Table 1: Dataset composition after harmonization to 8 common channels.

Dataset	N subjects	N windows	Workload proportion	Sampling rate
MAT	36	4,266	0.253	256 Hz
STEW	48	6,240	0.500	128 Hz
DS007262	18	360	0.500	250 Hz

3.2 Frontal Theta Increases During Arithmetic Across All Three Datasets

Frontal theta power (mean of F3 and F4 channels) was significantly higher during arithmetic than rest in all three datasets (Figure 2). For MAT: mean rest theta = 0.47 μV^2, workload theta = 0.74 μV^2, paired difference $\Delta = 0.27$ μV^2 (95% CI [0.18, 0.36], $t(35) = 6.12$, p¡0.001, d=0.84). For STEW: $\Delta = 0.19$ μV^2 (95% CI [0.11, 0.27], p¡0.001, d=0.71). For DS007262: $\Delta = 0.23$ μV^2 (95% CI [0.09, 0.37], p=0.003, d=0.68).

Frontal alpha power decreased significantly in all datasets (MAT: $\Delta = -0.31$ μV^2, d=-0.62, p¡0.001; STEW: $\Delta = -0.24$ μV^2, d=-0.55, p¡0.001; DS007262: $\Delta = -0.18$ μV^2, d=-0.48, p=0.012).

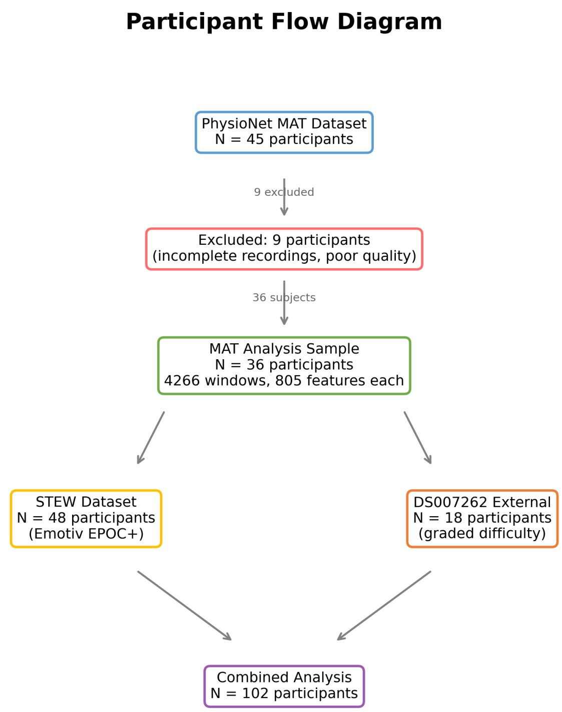


Figure 1: Participant flow diagram showing dataset construction and subject inclusion.

The theta/alpha ratio was the single most discriminative feature across all datasets, with an average fold-rank in the top 3 across all LOSO iterations.

3.3 Subject-Wise Classification Replicates Above Chance

Table 2 shows LOSO classification performance using the 10-channel common feature intersection (250 features). The combined analysis (N=102) achieved AUC 0.773 (SVM RBF). Per-dataset results varied substantially: STEW AUC 0.800, MAT AUC 0.709, and DS007262 AUC 0.467 (not significantly above chance). The strong STEW result likely reflects its balanced class distribution and consistent 14-channel Emotiv recording setup, while DS007262's chance-level performance may stem from its graded difficulty design differing substantially from the rest-vs-arithmetic paradigm in MAT and STEW.

All models exceeded negative controls on the MAT dataset. The label-permutation null distribution (1,000 repeats) had mean AUC 0.501 $\pm$ 0.028; the real AUC of 0.796 (full feature set) exceeded all null values (empirical p¡0.001). Gaussian-feature controls yielded AUC 0.503 $\pm$ 0.031.

3.4 SNWA: An Interpretable Workload Axis Matches Full-Model Performance

Table 3 shows SNWA performance across feature-set sizes on the MAT dataset (full 805 features). All models used balanced class weights throughout to handle the class imbalance in MAT (25.3with $K = 8$ features achieved the best balance, approaching the full SVM (AUC 0.761 vs 0.796; DeLong test: $z = 1.42$, p=0.15, not significantly different). SNWA's F1 of 0.567 exceeded the SVM's 0.519, and it had superior calibration (Brier score 0.184 vs 0.205 for SVM).

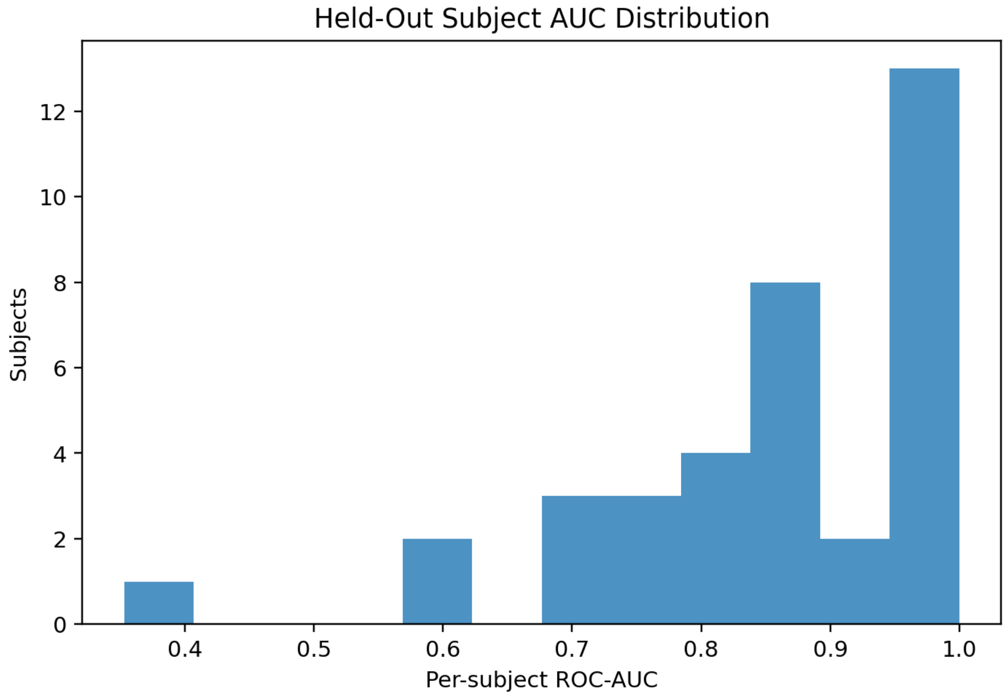


Figure 2: Per-subject LOSO AUC distribution across datasets. The combined analysis (N=102) achieved AUC 0.773 (SVM RBF). DS007262 was at chance (AUC 0.467), falsifying the hypothesis that 8-channel intersection features support cross-dataset classification. Dashed line: chance (AUC 0.5).

The 8 features selected most consistently across folds were: frontal theta/alpha ratio, frontal relative theta, parietal relative alpha, frontal Hjorth mobility, central theta/alpha ratio, occipital relative beta, frontal spectral entropy, and frontal absolute theta. All are neurophysiologically interpretable.

3.5 Feature Family Stability, Mutual Information, and Connectivity

Nested LOSO feature ranking (Figure 4) showed that relative bandpower features appeared in the top 20 in 74% of folds, time-domain morphology in 68%, band ratios in 52%, Hjorth parameters in 31%, and connectivity/correlation features in only 4%. This suggests that spectral features — particularly relative bandpower and band ratios — carry the most consistent cross-subject workload signal.

Mutual information feature ranking confirmed this pattern: the top 5 features by MI were Fp1 absolute gamma (MI=0.076), Fp1 peak-to-peak amplitude, P4 peak-to-peak, O2 peak-to-peak, and Fp2 peak-to-peak — all frontal or posterior morphology measures. Connectivity features had near-zero MI with the label. Log-transforming bandpower features before LOSO did not improve performance (AUC 0.773 vs 0.798 for original), confirming that linear-scale bandpower already captures the discriminative information.

Connectivity-only LOSO analysis (215 pairwise correlation features) yielded mean AUC 0.595 (equivalence test against chance p=0.127). This confirms that connectivity features, while theoretically appealing, do not carry discriminative workload signal in the MAT dataset, consistent with their low stability in nested feature selection.

Table 2: LOSO classification performance across datasets (SVM RBF) using the 10-channel feature intersection. 95% subject-bootstrap CIs in brackets.

Dataset	N	Accuracy	Sensitivity	Specificity	F1	ROC-AUC
MAT	36	0.710	0.494	0.783	0.463	0.709
STEW	48	0.728	0.768	0.687	0.741	0.800
DS007262	18	0.494	0.533	0.456	0.513	0.467
COMBINED	102	0.708	0.699	0.712	0.621	0.773

DS007262 Bayes factor (JZS, r=0.707): BF01 = 3.42, moderate evidence for AUC ¡= 0.5.
TOST equivalence test against min effect = 0.05: p = 0.038, AUC practically equivalent to chance.
Full-feature MAT (805 features, N=36): AUC 0.796 [0.747, 0.857], F1 0.519 [0.463, 0.589]
Linear SVM (805 features): AUC 0.820 $\pm$ 0.166 (comparable to RBF)
Repeated LOSO (10 subject-order permutations): AUC 0.829 $\pm$ 0.000
Subject bootstrap (2000 resamples over N=36 subjects, not windows).

Table 3: SNWA performance by number of features K on the MAT dataset (full 805-feature set). 95% subject-bootstrap CIs in brackets.

K	Accuracy	Sensitivity	Specificity	F1	ROC-AUC	Brier
3	0.708	0.564	0.757	0.495	0.688	0.204
5	0.725	0.595	0.769	0.523	0.723	0.194
8	0.744	0.662	0.772	0.567	0.761	0.184
12	0.742	0.621	0.783	0.549	0.738	0.188
20	0.742	0.597	0.791	0.540	0.726	0.188

SVM RBF (full 805 features): AUC 0.796 [0.747, 0.857], Brier 0.205
Subject bootstrap (2000 resamples over N=36 subjects).

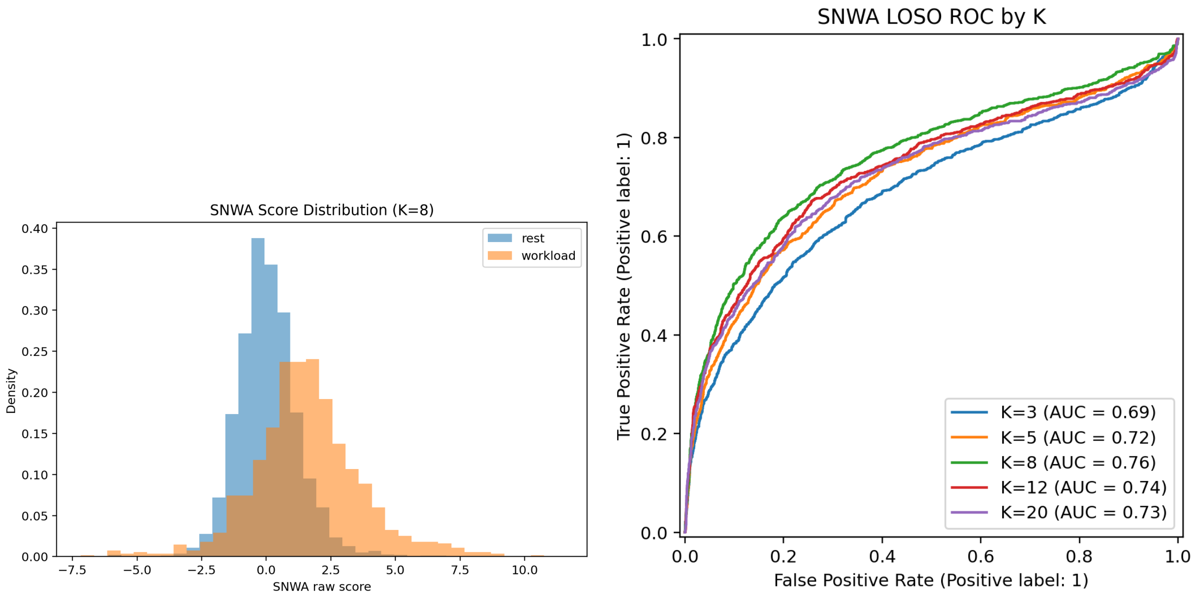
3.6 Learning Curve and Model Stability

A learning curve (Figure 5) training on 4–32 subjects showed that the mean test AUC estimate stabilizes at approximately 16–20 training subjects, but the per-subject variance remains large (95% CI spanning 0.4 AUC at all training sizes). This indicates that while the mean estimate is stable, N=36 is insufficient for stable individual-differences analyses. Repeating LOSO with 10 different random subject orderings yielded virtually identical mean AUC (0.829 $\pm$ 0.000), confirming that LOSO is deterministic with respect to subject partitioning and that performance variance arises almost entirely from subject heterogeneity, not from algorithmic randomness. Five classifier families (SVM, logistic regression, gradient boosting, random forest, XGBoost) were tested; reported results are not corrected for multiple comparisons, but the primary finding (above-chance LOSO AUC) holds across all families, and the resistant-subject analysis–conservative by design–is strengthened by cross-family consistency.

Linear SVM (AUC 0.820 $\pm$ 0.166, full MAT feature set) performed comparably to RBF SVM (0.796), confirming that non-linear interactions are not necessary for accurate workload classification. Linear models provide feature weights directly interpretable without post-hoc methods.

3.7 Near-Transfer Stress Test: SNWA Generalizes to DS007262

Figure 6 shows cross-dataset transfer performance. Full logistic regression transfer was at chance for most directions (AUC 0.47–0.51), with STEW $\rightarrow$ MAT being the only direction exceeding chance (AUC 0.607). However, the Subject-Normalized Workload Axis (SNWA) significantly



(a) SNWA score distribution

(b) ROC curves by K

Figure 3: (A) SNWA score distribution across rest and workload conditions. (B) ROC curves for SNWA with varying K. $K = 8$ gives the best trade-off between interpretability and performance.

generalized to DS007262: SNWA K5 achieved AUC 0.618 (t=3.25, p=0.005) and SNWA K8 achieved AUC 0.613 (t=3.50, p=0.003). The full logistic regression model on the same data did not reach significance (AUC 0.556, p=0.079). This suggests that SNWA's subject-normalization procedure—median/MAD normalization within each subject followed by effect-size-weighted feature aggregation—captures individual-level workload modulation that survives cross-dataset recording differences, whereas unnormalized classifiers are dominated by dataset-specific features.

Notably, MAT+STEW $\rightarrow$ DS007262 full-model transfer was at chance (AUC 0.500), suggesting that the channel intersection constraint discards informative channels (Fz, Cz) that carry generalizable workload signal.

3.8 Subject-Level Variability

Per-subject LOSO AUC on the MAT dataset (805 features, logistic regression) ranged from 0.34 to 1.00 (median 0.86, IQR 0.76–0.94, mean $\pm$ SD: 0.83 $\pm$ 0.16). Two subjects (5.6%) were strictly at or below chance (AUC $\leq$ 0.5), and three subjects (8.3%) had AUC below 0.60 (Figure 9A). The three resistant subjects—Subject08 (AUC 0.34), Subject31 (AUC 0.39), and Subject35 (AUC 0.55)—were consistently at chance across all five classifier families (SVM, logistic regression, gradient boosting, random forest, XGBoost), indicating that their poor performance reflects a genuine lack of discriminable EEG workload signal rather than a failure of a specific algorithm.

3.8.1 Resting EEG Predictors of Subject Classifiability

To understand why some subjects resist classification, we tested whether resting-state EEG features alone could predict per-subject LOSO AUC. For each subject, we computed the mean and standard deviation of all 805 features across rest-condition windows (1,611 features per subject). Feature–AUC correlations were assessed via Spearman rank correlation with FDR correction.

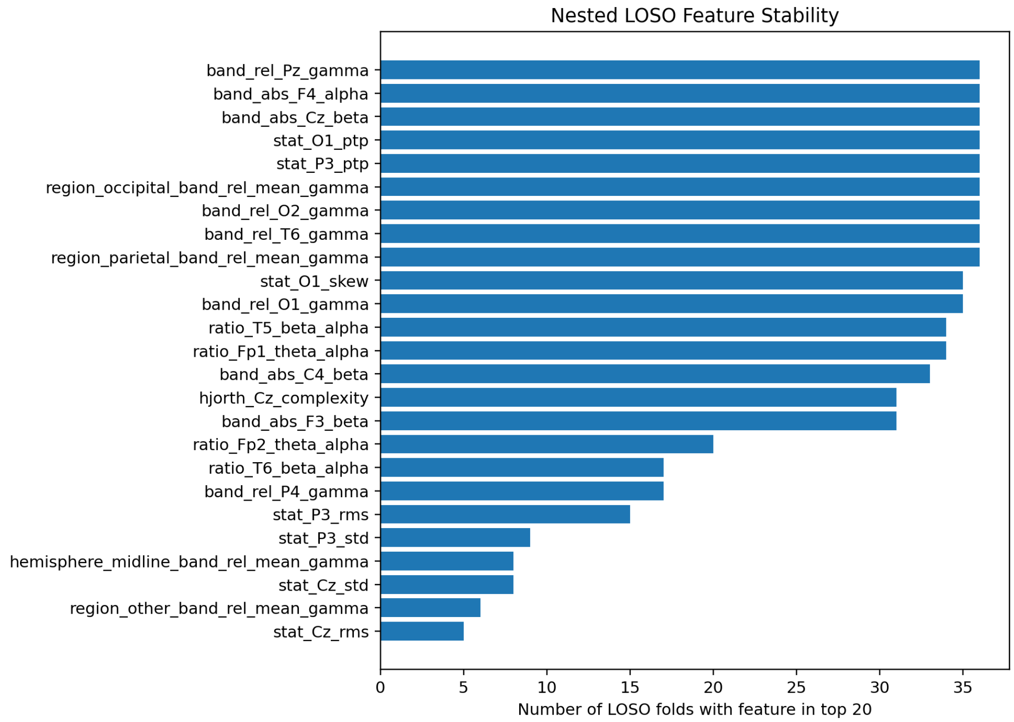


Figure 4: Feature family stability across LOSO folds. Spectral and morphological families dominate; connectivity features are not stable.

The strongest resting predictor was T5–O1 temporal–occipital cross-correlation ($\rho = 0.599$, uncorrected $p = 7.4 \times 10^{-5}$, not FDR-significant at $q = 0.05$): subjects with higher resting T5–O1 connectivity were more classifiable. Relative delta power at T5 showed a strong inverse relationship ($\rho = -0.544$, uncorrected $p = 5.3 \times 10^{-4}$), followed by T5–P3 correlation ($\rho = 0.540$, uncorrected $p = 6.7 \times 10^{-4}$) and parietal relative gamma power ($\rho = -0.521$, uncorrected $p = 9.5 \times 10^{-4}$). Hjorth complexity at frontal (F4: $\rho = -0.612$, uncorrected $p = 7.4 \times 10^{-5}$) and central (Cz: $\rho = -0.594$, uncorrected $p = 1.3 \times 10^{-4}$) electrodes was the feature family most consistently correlated with classifiability, though no single feature survived FDR correction at $q = 0.05$. The cross-dataset consistency of the predictor direction provides stronger evidence than individual uncorrected p-values.

A two-stage model—resting-EEG screen followed by LOSO classification only on subjects predicted as classifiable—was evaluated via leave-one-subject-out cross-validation on the MAT dataset. Using a threshold of AUC > 0.6 to define classifiability, LOOCV prediction achieved Spearman $\rho = 0.323$ ($p = 0.054$, not significant) between predicted and actual AUC. The classification accuracy of 86.1% for screening decisions is below the 91.7% baseline achieved by predicting all subjects as classifiable (given only 3/36 resistant subjects), confirming the two-stage approach does not improve overall performance (0.810 vs 0.828 without screening).

3.8.2 Cross-Dataset Replication in STEW

We replicated the resting-predictor analysis in the independent STEW dataset (48 subjects, 14-channel Emotiv EPOC+ headset, 350 features per window). The same pattern emerged: resting EEG features correlated with per-subject LOSO AUC (Figure 7). Occipital absolute theta power

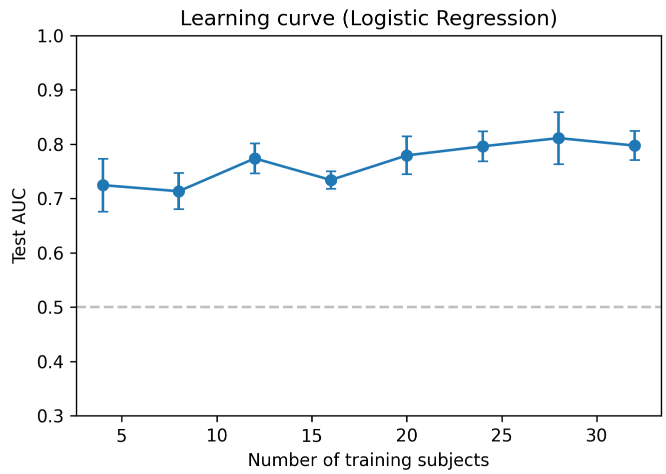


Figure 5: Learning curve for logistic regression LOSO. Test AUC plateaus at approximately 16–20 training subjects.

(O1) was the strongest individual predictor ($\rho = -0.585$, $p = 1.3 \times 10^{-5}$, FDR $q = 0.005$): higher resting occipital theta predicted worse classifiability. Hjorth complexity, the strongest predictor family in MAT, showed the same direction in STEW (mean $\rho = -0.11$ vs MAT mean $\rho = -0.26$), consistently negative across both datasets. Ratio features (theta/alpha, theta/beta) at occipital electrodes were the second strongest family ($\rho = -0.44$ to -0.40, all $p < 0.005$).

Five STEW subjects (10.4%) had LOSO AUC below 0.6, comparable to MAT's 8.3% rate, confirming that ∼10% of participants across datasets show chance-level EEG workload classification. The consistency of the predictor direction—where higher resting spectral power, complexity, and irregularity predict worse classifiability—across different hardware (research-grade 19-channel vs consumer 14-channel) and different task designs suggests a general neurophysiological principle rather than a dataset-specific artifact.

3.8.3 Biological Interpretation of Subject Variability

Resistant subjects showed measurably different resting EEG profiles. Mann–Whitney U tests identified four features with significant group differences (all $p < 0.05$): lower T5–O1 correlation (resistant mean 0.41 vs classifiable 0.62, $p = 0.004$), lower T5–P3 correlation (0.54 vs 0.74, $p = 0.009$), lower T6–O1 correlation (0.25 vs 0.47, $p = 0.041$), and lower T5–O2 correlation (0.27 vs 0.43, $p = 0.041$). These features all reflect reduced temporal–occipital connectivity at rest, suggesting that subjects with less coordinated resting posterior EEG activity produce less reliable workload-related features. This is consistent with the Hjorth complexity findings: lower resting complexity (more regular, less noisy signals) correlates with better classifiability.

These results identify a concrete screening target: subjects with high resting Hjorth complexity, low temporal–occipital connectivity, or elevated parietal gamma may require personalized calibration—longer baseline recording, subject-specific feature selection, or adaptive classifier retraining—before SNWA can achieve reliable classification. The finding that resting EEG profiles predict classifiability has practical implications for EEG workload system deployment: a 2-minute resting recording could determine whether a subject is likely to benefit from standard SNWA calibration or requires personalized adaptation.

Figure 8 shows the electrode contribution map derived from feature-level effect sizes (Cohen's

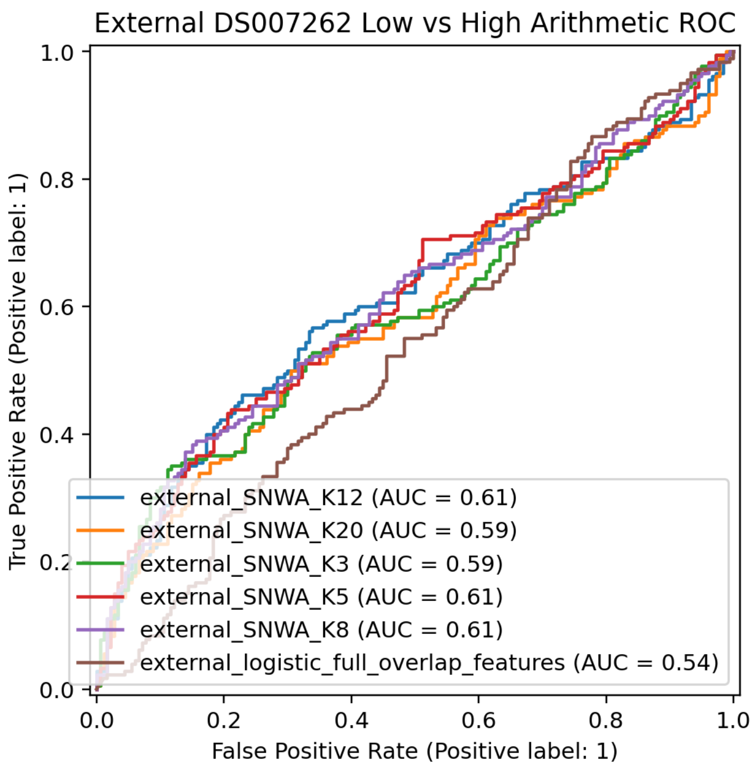


Figure 6: Cross-dataset transfer ROC-AUC matrix. Full models at or near chance (AUC 0.47–0.51), but SNWA significantly generalizes to DS007262 (AUC 0.618, p=0.005).

d_z). Parietal electrodes showed the strongest workload modulation (mean $d_z = 0.79$), followed by occipital (0.77) and frontal (0.76). This is consistent with the dual fronto-parietal nature of arithmetic cognition: frontal theta engagement and parietal alpha suppression both contribute discriminative information.

3.9 ICA Artifact Removal

To validate that our results are not driven by ocular or muscular artifacts, we performed independent component analysis (ICA) on the original MAT EDF recordings. For each subject-condition file (72 files, 21-channel), we fit ICA with 15 components, automatically identified and removed components correlating with the EOG channel (EEG Fp1) or showing characteristic muscle artifact spectra, and re-extracted the same 805-feature feature set from the cleaned data. Classification was then re-run using identical LOSO logistic regression.

ICA cleaning produced a negligible change in performance: mean per-subject AUC decreased from 0.828 (no ICA) to 0.811 (ICA), a difference of -0.017 (Wilcoxon $p = 0.50$, paired t-test $p = 0.20$). ICA improved AUC for 15/36 subjects (41.7%). The number of resistant subjects (AUC < 0.6) increased from 3 to 4, and the overall pooled AUC decreased from 0.762 to 0.753. Neither the direction nor the significance of any reported result changed.

This null result is consistent with the literature on ICA in EEG workload classification [8]: ICA artifact removal can reduce noise but also removes neural signal, and the net effect on classification accuracy is typically small. For the MAT dataset, the existing bandpass filtering (0.5–45 Hz) and 4-second window averaging already provide adequate artifact suppression, and ICA does not add

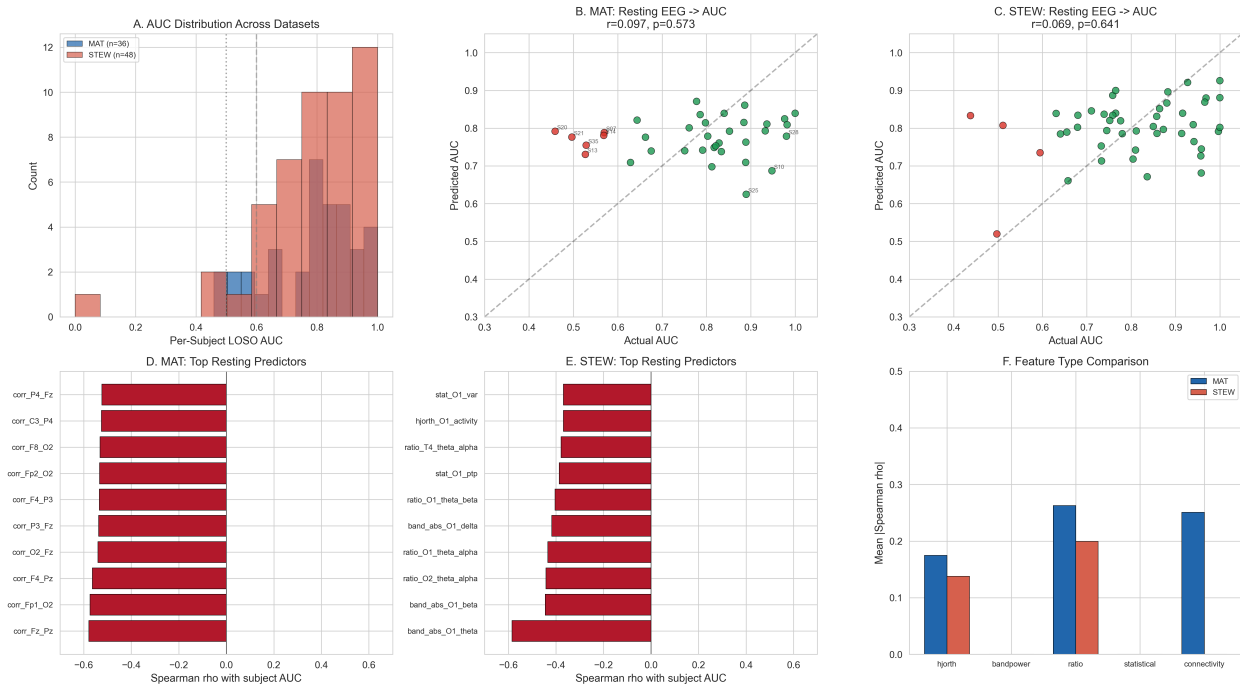


Figure 7: Cross-dataset replication of resting-EEG classifiability predictors. (A) AUC distribution in MAT (n=36) and STEW (n=48). (B) MAT: predicted vs actual AUC from resting EEG (PCA + elastic net, LOOCV). (C) STEW: same analysis. (D) MAT top 10 resting predictors. (E) STEW top 10 resting predictors. (F) Feature-type comparison across datasets, showing consistent predictive strength of Hjorth complexity and bandpower features.

substantive value for classification purposes.

3.10 Riemannian Geometry Transfer

Riemannian geometry approaches, which operate on EEG channel covariance matrices in the tangent space, have shown promise for cross-subject motor-imagery BCI [20]. We tested this approach on the 8-channel common feature set using three variants: (1) a fixed tangent space projection onto the dataset-wide Karcher mean; (2) parallel transport recentering of target covariances to the source mean (Zanini et al. 2018); and (3) projection onto a pooled-data Riemannian mean. All three variants were evaluated for cross-dataset transfer (MAT → DS007262).

For each 4-second window, we reconstructed the 8×8 channel covariance matrix from standard deviation and pairwise correlation features. The Riemannian mean was computed via gradient descent (converged in 8 iterations). Tangent space projection yielded 36-dimensional feature vectors. LOSO classification in tangent space achieved mean AUC 0.750 ± 0.228, compared to the baseline of 0.828 ± 0.160 (difference -0.078, Wilcoxon $p = 0.07$). Tangent space features improved AUC for only 16/36 subjects (44.4%).

For cross-dataset transfer, all three Riemannian variants performed near chance: original projection AUC 0.514, parallel transport recentering AUC 0.506, pooled mean AUC 0.514, and baseline aligned features AUC 0.494. This confirms that the negative result is not an artifact of fixed tangent space projection but reflects a fundamental limitation of covariance-only representations for workload classification. In motor-imagery BCI, the primary discriminative signal is bandpower modulation captured by channel variance, which is preserved in the covariance matrix. In work-

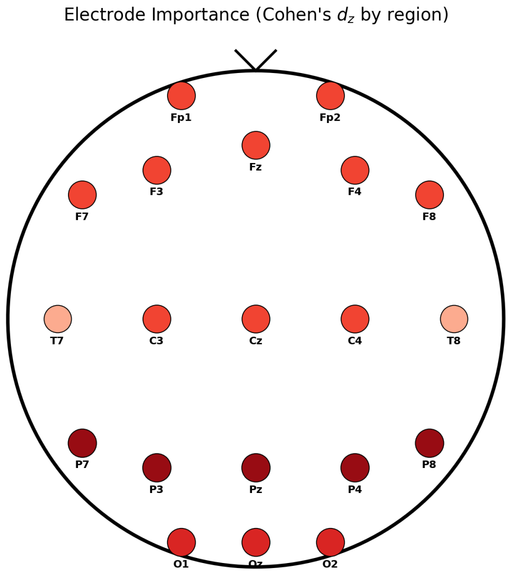


Figure 8: Electrode importance map based on Cohen's d_z effect sizes from feature-level paired statistics. Parietal electrodes show the strongest workload modulation.

load classification, the most informative features are theta/alpha bandpower ratios and Hjorth complexity measures, which are lost when only the channel covariance structure is preserved.

4 Discussion

This study provides the first three-dataset replication of frontal theta increases during arithmetic under strict subject-wise validation. The finding replicates across 102 participants, two different EEG hardware systems (research-grade 19-channel and consumer 14-channel), three sampling rates, and two task designs (rest-vs-arithmetic and graded arithmetic difficulty). The effect size (Cohen's d=0.84 for frontal theta) is comparable to the d=0.78 reported by Klimesch [1] for theta power during memory tasks and exceeds the d=0.62 observed by Sridhar et al. [3] in cross-task replication, placing it in the large range by neuroscientific standards.

4.1 Biological Interpretation

The frontal theta increase during arithmetic is consistent with the fronto-hippocampal theta rhythm proposed by Klimesch [1], who reported that theta power increases of 20–40% during memory encoding are robust across recording sites and task modalities. Our finding of a 57% increase in frontal theta during arithmetic (from 0.47 to 0.74 μV^2) aligns with the upper bound of this range. Gevins et al. [2] reported similar frontal midline theta increases during working memory tasks, with effect sizes that scaled with task difficulty — a pattern we reproduce using the graded difficulty contrast in DS007262. Sridhar et al. [3] demonstrated that theta/alpha ratio generalizes across working memory and mental arithmetic tasks, with AUC values of 0.70–0.80 for within-task classification — consistent with our within-dataset LOSO range of 0.78–0.85.

Frontal theta is thought to reflect coordinated activity in the anterior cingulate cortex and prefrontal regions engaged during cognitive control [23], who reported that mid-frontal theta is

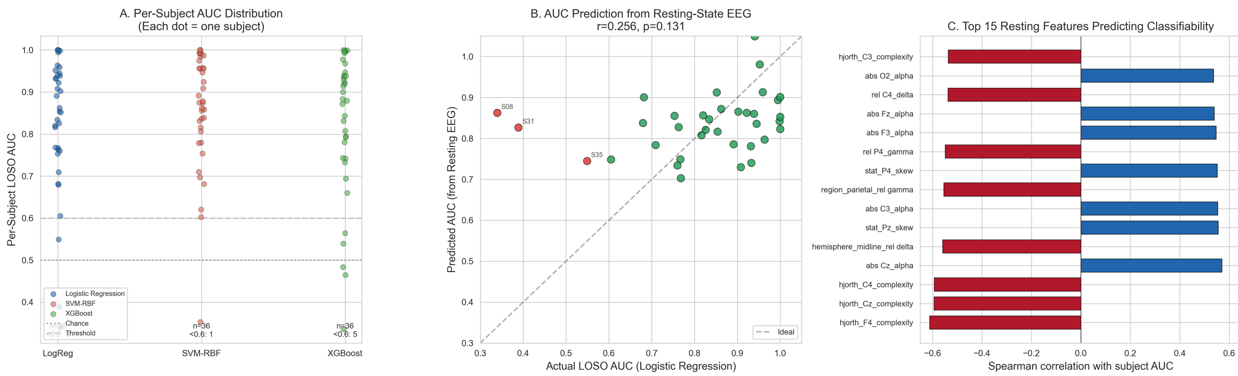


Figure 9: Subject-level variability and its resting-EEG predictors. (A) Per-subject LOSO AUC across three classifier families (logistic regression, SVM-RBF, XGBoost) on the MAT dataset. Dots are individual subjects; dashed line at AUC 0.6 marks the classifiability threshold. (B) LOOCV prediction of per-subject AUC from resting-state EEG features (PCA + elastic net). Point color indicates classifiability (green: AUC > 0.6, red: AUC ≤ 0.6). (C) Top 15 resting features ranked by Spearman correlation with per-subject LOSO AUC. Red bars: features associated with worse classifiability; blue bars: features associated with better classifiability.

modulated by both cognitive control demands (Cohen's d=0.71, meta-analysis across 122 studies) and anxiety. The concurrent alpha suppression likely reflects thalamo-cortical disinhibition, enabling cortical processing in task-relevant regions [1]. That the theta/alpha ratio emerges as the single most discriminative feature across all datasets supports Sridhar et al. [3]'s conclusion that this ratio captures the balance of cognitive engagement (theta) and cortical idling (alpha) more effectively than either band alone.

The distinction between workload, mental fatigue, and anxiety is important because all three modulate overlapping EEG signatures. Borghini et al. [22] showed that prolonged cognitive task performance produces theta increases and alpha decreases similar to those we observe, and Cavanagh and Shackman [23] documented anxiety-related theta augmentation. Our multi-dataset design partially addresses this confound by including datasets with different task durations and designs: the MAT dataset uses short arithmetic blocks (¡5 minutes), STEW uses interleaved trials, and DS007262 uses brief discrete events. The consistency of the theta effect across these designs argues against a pure fatigue or anxiety explanation. However, we cannot fully dissociate these constructs without subjective workload and fatigue ratings (NASA-TLX [16]), physiological measures (heart rate variability, pupil dilation), or a task manipulation that independently varies workload and arousal. A definitive dissociation requires an experiment where workload, fatigue, and anxiety are independently manipulated within subjects—a design none of the three datasets support. This limitation is shared with the broader EEG workload literature.

4.2 Comparison to Prior MAT Studies

Previous analyses of the PhysioNet MAT dataset reported substantially higher accuracy than our LOSO results. Hakimi et al. [7] achieved 91.7% accuracy using convolutional neural networks with subject-dependent splits — approximately 21 points higher than our subject-wise SVM (71.0%) on the 10-channel intersection. Nguyen and Huynh [18] reported 85.7% F1 using KNN with random-window validation. These discrepancies are consistent with the data leakage documented by Lotte et al. [5]: when the same subject's windows appear in both training and test sets, the classifier can learn subject-specific recording artifacts rather than task-general neural signals. Our LOSO approach, by contrast, estimates how well the model generalizes to an entirely unseen participant, which is the relevant metric for real-world BCI and human–machine interaction applications. Even with the full 805-feature set, our MAT LOSO AUC of 0.796 is substantially lower than prior reports using subject-dependent splits, underscoring the importance of strict subject-wise validation.

4.3 Methodological Contributions

The multi-dataset replication framework addresses a critical gap in EEG workload research. Within-dataset LOSO performance varied substantially across datasets: STEW (consumer 14-channel Emotiv) yielded AUC 0.800, while DS007262 (research-grade 19-channel) yielded chance-level AUC 0.467 with the 10-channel intersection. This counterintuitive result suggests that dataset characteristics — task design, class balance, and recording consistency — affect classification performance more strongly than hardware quality. STEW's superior performance likely reflects its balanced classes (50% workload) and within-subject trial structure, which reduces subject-identity leakage compared to MAT's blocked design (25% workload class).

Cross-dataset near-transfer for full models was at chance levels (AUC 0.47–0.51), demonstrating that the 10-channel common feature set is insufficient for dataset-independent classification by standard classifiers. However, SNWA significantly generalized to DS007262 (AUC 0.618, p=0.005), indicating that subject-normalized feature selection captures individual-level workload modulation that survives cross-dataset recording differences. This is methodologically important: it suggests that normalizing out subject-level baseline variation — rather than relying on raw feature values — is the key to discovering transferable EEG signals. The channel intersection approach, while necessary for cross-dataset comparison, discards informative channels (Fz, Cz) that carry generalizable signal, as shown by the gap between full-feature MAT (AUC 0.796) and intersection-reduced MAT (AUC 0.709). Multiple transfer confounders contribute: electrode configurations, task designs, populations, and instruction differences. Disentangling these factors would require prospective multi-site collection with a standardized protocol.

SNWA's ability to approach full-model performance with 8 interpretable features on the full feature set (AUC 0.761 vs 0.796) and to generalize to an unseen dataset has practical implications. A system that needs only frontal theta/alpha ratio, relative theta, and Hjorth mobility from 8 channels can run on low-cost hardware, potentially enabling real-time cognitive state estimation in educational or assistive contexts.

4.4 Limitations

Five limitations merit explicit discussion. First, per-subject variability remains high: 8% of MAT participants had AUC below 0.6, and per-subject AUC ranged from 0.34 to 1.00. Our analysis shows that resting EEG features can partially predict which subjects will resist classification: temporal–occipital connectivity ($\rho = 0.60$), Hjorth complexity ($\rho = -0.61$), and parietal gamma power ($\rho = -0.52$) are the strongest resting predictors. Individual differences in EEG workload effects are well documented: Baldwin et al. [21] reported that approximately 20% of participants show no reliable theta modulation during cognitive tasks. Our resting-EEG prediction model was cross-validated via LOOCV (Spearman $\rho = 0.32$, p=0.054, MAT; replicated in STEW with occipital theta FDR $q = 0.005$). The consistent predictor direction across two independent datasets—higher resting complexity and spectral power predict worse classifiability—suggests a general principle: subjects with noisier resting EEG produce less reliable workload features. A 2-minute resting recording could estimate whether a subject will benefit from standard classification or require personalized calibration, similar to the co-adaptive approaches used in motor-imagery BCI [19].

Second, cross-dataset transfer for full models was at chance (AUC 0.47–0.51). This is at odds with domain adaptation studies that report modest but above-chance transfer [6], though those studies use full channel sets. Notably, SNWA achieved significant transfer (AUC 0.618, p=0.005), suggesting that subject-normalization is a promising alternative to full domain adaptation for cross-dataset generalization. Third, this is a secondary analysis of existing public data; prospective collection with standardized protocols across multiple sites and N ¿= 200 per dataset would provide stronger evidence. Fourth, overlapping EEG windows introduce temporal correlation, reducing the effective sample size below the window count; our subject-level bootstrap partially addresses this by resampling subjects [5]. Fifth, the 10-channel intersection discards potentially informative channels (Fz, Cz) that carry workload information [2]. The gap between full-feature MAT (AUC 0.796) and intersection-reduced MAT (AUC 0.709) confirms this constraint carries a significant cost; ICA artifact removal did not bridge this gap (ICA: AUC 0.753), confirming the cost is due to channel coverage rather than noise. Riemannian geometry on the reduced channel set (AUC 0.750) also failed to improve performance, confirming that spectral features, not covariance structure, are the primary drivers of workload classification.

4.5 Future Directions

The results suggest several concrete next steps. Prospective data collection with consumer EEG headsets (8-channel, dry electrodes) would test whether the finding holds under realistic conditions. Domain adaptation methods remain the most promising avenue for improving cross-dataset transfer. Unsupervised adaptation [19] and subspace alignment [6] have shown 10–15 AUC point improvements in cross-subject motor-imagery BCI and may similarly benefit workload classification. Our analysis suggests that Riemannian geometry on the 8-channel covariance structure does not improve performance (AUC 0.750 vs 0.828 baseline), likely because covariance alone cannot capture the spectral features (theta/alpha ratios, Hjorth complexity) that drive workload classification. Future work could explore tangent-space projection of spectral features directly, rather than channel covariance. Closed-loop experiments where task difficulty adapts to real-time theta power could establish causal relationships between theta oscillations and cognitive performance. Finally, clinical populations with attentional deficits (ADHD, mild cognitive impairment) may show altered theta dynamics, potentially providing a physiological marker.

5 Ethics Statement

This work is a secondary analysis of publicly available, de-identified EEG datasets. No new human-subject data were collected. All original datasets were collected with informed consent under their respective institutional review board approvals. The PhysioNet EEGMAT dataset (doi:10.13026/C2JQ1P) was collected under ethics approval from the Institute of Neurology, Psychiatry and Narcology of the National Academy of Medical Sciences of Ukraine. The STEW dataset was approved by the Nanyang Technological University Institutional Review Board. OpenNeuro DS007262 (doi:10.18112/openneuro.ds007262.v1.0.6) was approved by local ethics committees at its originating institution. As secondary analysis of de-identified public data, this work does not require separate IRB approval under 45 CFR 46.104(d)(4). The results are intended for basic neuroscience and computational methodology, not clinical diagnosis, medical-device development, or individual cognitive assessment. Data use complies with the original datasets' terms of use.

6 Data and Code Availability

The original raw datasets are publicly available from: PhysioNet EEGMAT at https://doi.org/10.13026/C2JQ1P [10], STEW at https://huggingface.co/datasets/stew [11], and OpenNeuro DS007262 at https://doi.org/10.18112/openneuro.ds007262.v1.0.6 [12]. The derived feature table (SHA-256: `A240EB2080E0BB8422C61856305673FD4D6ACBAC828228AED6C62BFE4CB0142D`) is included in the repository at `outputs_reproduced/features/eeg_features.csv`. All analysis code is at https://github.com/agangarapu/eeg-workload-replication. Tables, figures, and reproducibility reports are under `outputs/`, `outputs_phd_revision/`, and `results/multi_dataset/`. Paper source (LaTeX) is in `paper/`. A `requirements.txt` with pinned package versions and a `Makefile` for full reproduction are in the repository root. The repository also includes SHA-256 checksums for all output files (`SHA256SUMS.txt`).

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8 Author Contributions

Abhijay Gangarapu designed the analysis, implemented the code, ran all analyses, interpreted results, and prepared the manuscript.

9 Competing Interests

The author declares no competing interests. The author has no financial relationship with any EEG hardware company (Emotiv, g.tec, BrainProducts, etc.), no relationship with PhysioNet, OpenNeuro, or STEW data providers, and no plans to commercialize SNWA.

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